Part 1 — Analysis

Tools. DuckDB SQL for every metric (`queries.sql` has each statement in execution order), pandas for pivots and an independent cross-check, matplotlib for charts. `python run_analysis.py` regenerates this file, the tables and the charts from `data/raw/`.

Definitions applied exactly as the brief states. A *customer* is a user with at least one **delivered** order, counted under their `acquisition_channel`. CAC = total channel spend ÷ customers from that channel. Revenue per customer = delivered revenue ÷ customers (Jan–Jun). Repeat rate = % of customers with ≥ 2 delivered orders. LTV = revenue per customer × 30%. LTV:CAC = LTV ÷ CAC. Money is rounded to the nearest rupee and percentages to one decimal **only at display time**.

Population. After cleaning, 2,678 users have a delivered order: 2,603 across the five channels, 40 with no channel, 35 internal test accounts. Channel metrics use the 2,603.

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1. Data quality

Four issues were found. Rows are flagged, not deleted, so every exclusion is visible and reversible (`src/queries.py`, clean layer).

#	Issue	Rows affected	How handled
1	Exact duplicate rows in orders.csv (same order_id, every column identical)	206 of 5,684 rows (3.6%); 5,478 unique orders remain	Dropped with `SELECT DISTINCT` before any count. 0 order_ids had conflicting values, so this is a re-export, not two versions of an order.
2	Blank `acquisition_channel` in users.csv	72 users (1.4%); 81 orders, 40 customers, ₹85,089 delivered revenue	Cannot be attributed to a channel. Excluded from every channel metric and shown as an 'Unknown channel' row in the Q6 reconciliation so totals still tie.
3	Internal / test accounts: email on the company's own domain `@urbancart.in`	35 users, **all tagged Affiliate**; 285 delivered orders (6–11 per account, avg order ₹2,736 vs ₹1,169 overall)	Excluded from all metrics. This is the issue that matters most: with these accounts in, Affiliate's LTV:CAC is 1.10; without them it is 0.42 (table below).
4	Cancelled and returned orders	315 cancelled + 364 returned of 5,478 unique orders (12.4%)	Not 'successful' per the brief. Excluded from revenue, customer counts and repeat rate. Kept for the cancel/return-rate check in the supporting tables.

Why issue 3 matters — Affiliate with and without the 35 internal accounts

Affiliate metrics	Customers	Revenue per customer	Repeat rate	CAC	LTV:CAC
with internal accounts (naive)	540	₹2,339	14.1%	₹641	1.10
without internal accounts (clean)	505	₹957	8.1%	₹685	0.42

The 35 accounts place 285 large, always-delivered orders between them. Left in, they make Affiliate look like the second-best channel. Removed, Affiliate is the worst on every measure.

Checks that passed (run so the reader knows they were looked for):

- orphan orders (user_id not in users.csv): **0**
- orders dated before the user's signup: **0**
- null dates / values / status in orders: **0**; null fields in users: **0**
- zero or negative order values: **0** (range ₹199–₹3,991, no extreme outliers)
- duplicate user_ids: **0**; duplicate emails: **0**
- casing / whitespace variants: none — 3 status values, 5 channel values, all canonical
- spend table complete: **24 rows** = 4 channels × 6 months, total ₹15,90,000 (the brief's 'about ₹16 lakh')
- signups or orders outside Jan–Jun 2026: **0**
- users who never ordered: 2,097 (41%) — normal, not an error

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2. Acquisition cost (paid channels, Jan–Jun 2026)

Channel	Total spend	New customers	CAC
Google Ads	₹4,00,000	655	₹611
Affiliate	₹3,46,000	505	₹685
Meta Ads	₹5,47,000	675	₹810
Influencer	₹2,97,000	307	₹967

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3. Customer value (all channels, including Organic)

Channel	Customers	Delivered revenue	Revenue per customer	Repeat purchase rate
Influencer	307	₹8,77,469	₹2,858	73.0%
Google Ads	655	₹15,13,880	₹2,311	57.7%
Organic	461	₹8,77,167	₹1,903	55.7%
Meta Ads	675	₹11,24,892	₹1,667	44.7%
Affiliate	505	₹4,83,387	₹957	8.1%

Organic customers (461) carry no spend, so they appear here but not in CAC or LTV:CAC.

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4. Return on spend

Channel	Revenue per customer	LTV (× 30%)	CAC	LTV:CAC
Google Ads	₹2,311	₹693	₹611	1.14
Influencer	₹2,858	₹857	₹967	0.89

Channel	Revenue per customer	LTV (× 30%)	CAC	LTV:CAC
Meta Ads	₹1,667	₹500	₹810	0.62
Affiliate	₹957	₹287	₹685	0.42

Best: Google Ads at 1.14. Worst: Affiliate at 0.42. Only Google Ads returns more contribution than it costs. Note that "LTV" here is six months of revenue × 30%, so every figure is a floor; the *ranking* is what the recommendation rests on.

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5. Trend — CAC by month (new customers grouped by signup month)

Channel	Jan	Feb	Mar	Apr	May	Jun
Google Ads	₹592	₹621	₹596	₹602	₹644	₹611
Meta Ads	₹491	₹618	₹789	₹886	₹958	₹1,115
Affiliate	₹709	₹709	₹728	₹633	₹667	₹682
Influencer	₹981	₹980	₹913	₹960	₹964	₹1,000

Change from January to June:

Channel	CAC Jan	CAC Jun	Change (₹)	Change (%)	Spend change Jan→Jun (%)
Meta Ads	₹491	₹1,115	+₹624	+127.2%	+114.8%
Google Ads	₹592	₹611	+₹19	+3.2%	-25.7%
Influencer	₹981	₹1,000	+₹19	+2.0%	+2.0%
Affiliate	₹709	₹682	-₹27	-3.9%	-26.2%

Meta Ads is getting more expensive: ₹491 in January to ₹1,115 in June, +₹624 (+127%). Its monthly spend rose +115% (₹54,000 → ₹1,16,000) while new customers stayed flat (110 → 104), the signature of a saturated channel. Every other channel moved within a few percent (Google Ads +3.2%; Influencer +2.0%; Affiliate -3.9%).

Cohort caveat. June signups had at most 30 days to place an order, so June CAC is slightly overstated for every channel and the June repeat rate collapses (below). Meta's trend holds without June: January to May is ₹491 → ₹958 (+95%).

Signup month	Customers	Repeat rate
2026-01	456	55.5%
2026-02	432	57.4%
2026-03	401	53.6%
2026-04	481	54.3%
2026-05	463	43.8%
2026-06	370	5.9%

Supporting monthly detail — spend and new customers by signup month:

Channel	Jan	Feb	Mar	Apr	May	Jun
Google Ads	₹74,000	₹59,000	₹62,000	₹74,000	₹76,000	₹55,000
Meta Ads	₹54,000	₹68,000	₹86,000	₹1,09,000	₹1,14,000	₹1,16,000
Affiliate	₹61,000	₹56,000	₹59,000	₹69,000	₹56,000	₹45,000
Influencer	₹51,000	₹50,000	₹42,000	₹48,000	₹54,000	₹52,000

Channel	Jan	Feb	Mar	Apr	May	Jun
Google Ads	125	95	104	123	118	90
Meta Ads	110	110	109	123	119	104
Affiliate	86	79	81	109	84	66
Influencer	52	51	46	50	56	52

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6. Sanity check — Google Ads CAC = ₹611

I verified this number four independent ways:

1. **Reconciliation.** Every user with a delivered order must land in exactly one bucket: 505 + 655 + 307 + 675 + 461 + 40 (unknown) + 35 (internal) = 2,678 = the 2,678 distinct customers in the deduped orders table. It ties.

Bucket	Customers
Affiliate	505
Google Ads	655
Influencer	307
Meta Ads	675
Organic	461
internal / test accounts	35
unknown channel	40
total distinct customers (deduped orders)	2,678

2. **Hand arithmetic.** Google spend = 74k + 59k + 62k + 74k + 76k + 55k = ₹4,00,000; ₹4,00,000 ÷ 655 = 610.69 → ₹611.
3. **Second tool.** Recomputed in pandas with no SQL (`pandas_crosscheck` in `src/analysis.py`): 655 customers, CAC 610.69. Identical to the DuckDB result to the paisa; `run_analysis.py` asserts this on every run, and also asserts that the CAC used inside the LTV:CAC query equals the Q2 CAC (that assertion caught a fan-out join bug during development).
4. **Plausibility.** 655 customers from 1,100 Google signups is a 59.5% signup-to-customer rate, in line with the other channels (table below), and a ₹611 CAC is the cheapest of the four, consistent with intent-driven search traffic.

Supporting tables

Cancel/return rate by channel (clean users) — a second, independent signal on Affiliate traffic quality:

Channel	Orders	Cancelled or returned
Affiliate	913	40.2%
Meta Ads	1,192	9.7%
Google Ads	1,293	6.8%
Organic	892	6.3%
Influencer	822	5.7%

Signup → customer conversion by channel:

Channel	Signups	Customers	Signup → customer
Google Ads	1,100	655	59.5%
Organic	828	461	55.7%
Meta Ads	1,300	675	51.9%
Influencer	600	307	51.2%
Affiliate	1,200	505	42.1%

Spend by quarter (the baseline used for the 20% cut in Part 2):

Channel	Q1 (Jan–Mar)	Q2 (Apr–Jun)	H1 total
Meta Ads	₹2,08,000	₹3,39,000	₹5,47,000
Google Ads	₹1,95,000	₹2,05,000	₹4,00,000
Affiliate	₹1,76,000	₹1,70,000	₹3,46,000
Influencer	₹1,43,000	₹1,54,000	₹2,97,000

Assumptions

- A customer belongs to their signup `acquisition_channel` for the whole period (last-touch at signup), as the brief defines.
- Overall CAC uses Jan–Jun totals; the monthly CAC in Q5 matches each month's spend to customers who *signed up* that month, as instructed.
- Revenue is the `order_value` of delivered orders; no discount, shipping or COGS data is available, hence the brief's flat 30% margin.
- `@urbancart.in` emails are treated as internal/test traffic. Had they been genuine, they would still be 35 accounts generating 285 orders at 2.3× the average basket — an outlier population that should not drive a budget decision.
- Unknown-channel customers (40) are excluded rather than redistributed; spreading them pro-rata would not change any ranking.

- The six-month window truncates LTV for late cohorts; all LTV figures are floors and are compared only against each other.